

SOVEREIGN COGNITION / RELEASE HANDOFF

Project Intermix

Cognition v1.4.0

Exact numbers. Typed second-brain memory. Temporal grounding. Bounded 8K creation.

TARGET Google Pixel 10 Pro / Tensor G5 / Android 17 / Termux	ENGINE Gemma 4 E4B through LiteRT-LM 0.16.1 on OpenCL GPU
PHYSICAL CONTEXT 8,000-token KV cache with mode-specific input/output partitions	MEMORY SQLite schema v2 with FTS5, typed events, revisions, watches, and reports
AUTONOMY Fixed workspace read/write/test; reviewed deletion; app-open-and-idle maintenance	VOICE Resonance v1.3.1 preserved as optional and offline-safe

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This handoff describes the installed architecture, operator controls, privacy model, device validation, rollback path, and deferred Observatory experiment.

01 / EXECUTIVE BRIEF

What Cognition changes

Cognition increases effective continuity and factual integrity without increasing the physical mobile context allocation.

NUMBER INTEGRITY Controller-owned anchors protect exact times, dates, versions, percentages, currency, and arithmetic.	SECOND BRAIN Typed work, project, goal, preference, commitment, and private wellbeing timelines are independently inspectable.
TEMPORAL ROUTING Stable topics stay local; evolving tech and current-world questions search under a deterministic policy.	PROVIDER WAVES Optional APIs and official specialist endpoints run within six total requests and three-way concurrency.
CREATION STUDIO The /create mode reserves 4K output tokens for scripts and lore while keeping the total at 8K.	IDLE REPORTS Inference-free Markdown/PDF snapshots document trajectory without warming Gemma.

Core design rule

The model writes language. The controller owns release identity, clock, numeric truth, prompt budgets, tool results, memory policy, evidence acceptance, and completion state.

Preserved foundation

- Resident LiteRT Engine with fresh conversations for assembled virtual contexts.
- SQLite transcript continuity and session rehydration.
- Ordered workspace repair epochs with real post-write tests.
- Workspace Lens, atomic checkpoints, deletion review, cancellation, and recoverable trash.
- Clarity Markdown finalization and restrained cyan-purple terminal hierarchy.
- Optional Resonance bridge; this release does not start Debian, load Kokoro, or alter WAV files.

02 / RUNTIME TOPOLOGY

Controller-centered architecture

Every generated answer passes through deterministic selection, validation, persistence, and display layers.

STEP	LAYER	ROLE
1	User / TUI	Input, explicit slash mode, cancellation, file review
2	Intent and integrity	Response policy, freshness decision, exact-number ledger, local/UTC clock
3	Context assembly	Mandatory truth blocks plus ranked memory, mission, project, and evidence blocks
4	Gemma 4 E4B	Fresh conversation on resident 8K LiteRT Engine; language or bounded tool proposal
5	Protocol and guards	Hide control tags, verify citations, validate numbers, execute ordered tools
6	Commit and render	SQLite message/event records, task state, final Markdown, optional completed-answer voice

Persistent stores

STORE	LOCATION	CONTENTS
Core memory	~/project-intermix/memory/sovereign.db	Sessions, messages, memories, typed events, watches, reports, web cache
Workspace	~/storage/downloads/sovereign_workspace/	User files, reports, project index, checkpoints, mission metadata
Rollback	~/project-intermix/archive/pre_cognition_install_*	Complete pre-install engine and memory copy plus version/launcher
Model	~/project-intermix/models/	Gemma E4B model and compiled runtime caches; not copied into release ZIP

Failure behavior

- Insufficient web evidence stops a forced current answer instead of permitting speculation.
- A failed workspace tool stops later generated actions and starts a fresh repair epoch from verified feedback.
- A still-invalid numeric repair displays controller anchors rather than the corrupted draft.
- Low storage skips idle network/report work; user activity cancels active maintenance.

03 / VIRTUAL CONTEXT

An 8K window that feels longer

Continuity comes from selective retrieval and reset-safe state, not from keeping every old token resident.

MODE	INPUT	OUTPUT	BEHAVIOR
Brief	7,200	800	Up to five sentences
Normal	6,800	1,200	Up to ten sentences unless complexity requires more
Deep	5,800	2,200	Detailed structured explanation
Workspace	5,200	2,800	Tool actions, file content, verification, repair
Create	4,000	4,000	Long-form script, lore, treatment, or creative artifact

Mandatory, non-evictable blocks

- Installed identity and system contract.
- Current user request and selected response/tool policy.
- Controller clock plus exact numeric anchors.
- Accepted web evidence and citation contract when grounding is active.
- Verified mission ledger in workspace mode.

Ranked and compactable blocks

- Recent turns, session checkpoint, semantic memories, typed timeline events, project history, and web background.
- Selection uses relevance, recency, salience, domain, pin state, and hard per-block token ceilings.
- Raw transcripts and mission records remain in SQLite after leaving the prompt.

Creation Studio

Use /create for high-lore scripts and video writing. It does not enlarge the KV cache or keep an ever-growing generation session; it reallocates half of the existing window to output and asks for a complete artifact rather than process commentary.

04 / NUMERIC INTEGRITY

Exact anchors before interpretation

The number guard was added after Gemma transformed 17:17 into unrelated repeated digits and faulty arithmetic.

PHASE	DETERMINISTIC ACTION
Extract	Capture ISO dates, clock times, versions, percentages, currency, and explicit numeric tokens from the current prompt.
Normalize	Store raw form, normalized form, digits, digit sum, and digital root. Inject exact local and UTC controller clocks.
Generate	Buffer number-sensitive conversational output instead of streaming it directly to the screen.
Validate	Require prompt anchors to survive and recompute written arithmetic using a safe Decimal/AST evaluator.
Repair	Run one private repair epoch with violations and the immutable ledger; discard hidden memory proposals from a bad draft.
Fail closed	If repair fails, show the exact anchors and verified arithmetic in a deterministic fallback.

Reference test

17:17 on the clock. Explain its numerology and show the exact digit sum.

Expected controller facts

- Anchor: 17:17 everywhere.
- Digit sum: $1 + 7 + 1 + 7 = 16$.
- Digital root: $1 + 6 = 7$.
- Any spiritual/numerological meaning is explicitly interpretive, not verified external fact.

Tradeoff

Strict numeric prompts reveal their answer only after validation, so the first visible text may arrive later.
Ordinary non-numeric conversation keeps the existing streaming path.

05 / MEMORY MATRIX V2

A typed, inspectable second brain

The new schema preserves old sessions while adding distinct timelines, provenance-bearing watches, and report records.

STRATUM	PURPOSE	RETRIEVAL
Messages	Immutable conversation and tool transcript	Recent turns, session view, FTS
Session checkpoint	Summary, task, decisions, open loops, tone, project	Always for active session within budget
Memories	Durable facts/preferences with revision history	FTS, relevance, salience, recency, pins
Memory events	Work, project, goal, commitment, preference, wellbeing timeline	Domain and current-query match
Project events	Verified file actions, hashes, exit codes, outputs	Prior-work and workspace context
Fact watches	Volatile queries, cadence, sources, status	Due idle refresh or explicit view
Report runs	Snapshot hash, Markdown/PDF path, privacy mode	Report commands and file explorer

New additive objects

- memory_events plus FTS5 index/triggers
- memory_entities and memory_relations for future lightweight knowledge graphing
- fact_watchlist for bounded temporal refresh
- report_runs for deterministic snapshot history

Conflict policy

Changed durable values preserve revision history. Volatile hard facts require matching web provenance. A transient self-report is not silently promoted into an identity fact, and model-generated preference/lore is never treated as the user's preference.

06 / SENSITIVE CONTINUITY

Private wellbeing memory with explicit consent controls

Cognition can remember relevant mental-health context while minimizing inference, irrelevant recall, and report leakage.

CAPTURE Explicit first-person feeling, named diagnosis, medication, or sleep statements only.	NO INFERENCE No sentiment-derived diagnosis, third-party health inference, or metaphor-to-pathology conversion.
QUERY GATE Private entries enter prompts only when the new request itself concerns wellbeing.	RETENTION Default 365-day expiry for new events; configurable from 1 to 3,650 days.
CONTROL Inspect, disable new capture, forget, or physically purge individual event IDs.	REPORTS Private text omitted by default; explicit report-level opt-in is separate from capture.

Operator commands

```
/memory audit
/memory sensitive status
/memory sensitive on|off
/memory sensitive retention 365
/memory timeline wellbeing
/memory event forget <id>
/memory event purge <id>
/report sensitive on|off
```

Storage disclosure

Intermix does not encrypt SQLite. Its primary protection is Android/Termux app-private storage. Exports and reports inherit the permissions of their destination. This is a continuity tool, not a clinical record, diagnostic model, crisis service, or replacement for professional care.

07 / TEMPORAL GROUNDING

Search only when knowledge can realistically be stale

A deterministic policy makes the web decision before Gemma can rationalize one.

RISK LANE	DEFAULT	EXAMPLES
Stable	Local	Math, theorem, timeless background
Interpretive	Local	Spirituality, symbolism, numerology meaning
Creative	Local	Script, lore, rewrite, brainstorming
Local task	Local	Long supplied design brief or code/document task
Volatile tech	Search	Current API, package, driver, framework, release
Current world	Search	News, price, law, election, schedule, score, weather

Bounded search waves

WAVE	PROVIDERS	WHEN
Configured	Brave, Tavily, Exa, SearXNG	Only when environment configuration exists
Specialist	GitHub REST, PyPI JSON, Crossref, arXiv	Query-specific official/package/research terms
Discovery	Google News RSS, Bing News RSS, Wikipedia	Current news or stable background
Fallback	DuckDuckGo HTML then Lite	Sequential final route with shared cooldown

Hard envelope

- Six requests maximum; three concurrent; result URL deduplication.
- Subject qualifier, intent, source age, and authority-aware evidence gates.
- Exponential provider cooldown to avoid repeated rate-limit hammering.
- Forced web output remains hidden until it cites accepted source IDs.

Refresh, document, and report without stealing the session

Maintenance is bounded, cancellable, storage-aware, and active only while the cockpit is open and idle.

PHASE	ACTION	LIMIT
Documentation	Hash changed workspace files and update project structure	Changed files only
Freshness	Refresh due volatile evidence with provenance	At most 2 watches per pass
Reporting	Create Markdown and minimal reliable PDF snapshot	Once daily unless forced; state hash
Research	Inspect configured trusted repository/registry/community leads	Six-hour default interval

Cancellation and pressure guards

- Any user typing or workspace activity cancels maintenance.
- Below 8 GiB free storage, temporal refresh and research/report work skip.
- No idle report loads Gemma; output is constructed from verified SQLite and workspace state.
- Private wellbeing content is excluded unless report consent is separately enabled.

Report location

```
~/storage/downloads/sovereign_workspace/reports/  
Intermix_Cognitive_Snapshot_YYYYMMDD.md  
Intermix_Cognitive_Snapshot_YYYYMMDD.pdf
```

Why PDF is controller-owned

A status handoff must preserve exact release, paths, counts, mission results, and privacy state. Deterministic rendering is faster, cheaper, and less likely to hallucinate than asking Gemma to narrate its own system state while idle.

Collaboration without blind trust

The Core may propose and execute bounded work, but verified state remains visible and recoverable.

CAPABILITY	DEFAULT	GUARD
Read	Automatic	Fixed workspace path; reserved control state excluded
Write	Automatic	Atomic replacement plus content-addressed checkpoint
Run Python	Automatic	Bounded runtime, captured stdout/stderr, cancellation
Repair	Automatic epochs	Failure stops remaining speculative actions
Complete	Controller-owned	Successful post-write test required
Delete	Review required	Hash revalidation and recoverable trash
Research	App-open and idle	Trusted configured sources; fetched text never executes

Cockpit hierarchy

- Cyan identifies active system functions; purple identifies memory/navigation; white/grey carries generated text.
- Streaming uses a throttled plain surface; final output receives one Markdown render.
- Long diagnostics are foldable; warnings remain visible.
- Ctrl+E opens Workspace Lens, Ctrl+S saves, Ctrl+X cancels work, Ctrl+V opens the voice archive.
- Resonance exposes online/offline state and only receives a completed, filtered answer after user action.

Terminal limitation

Textual cannot create true per-widget font sizes inside terminal cells. Professional hierarchy therefore comes from weight, color, spacing, borders, collapsible regions, width control, and final Markdown structure rather than browser-style typography.

10 / OPERATOR QUICK REFERENCE

Commands that expose every important decision

No hidden memory or web policy should require trusting a narrative answer from the model.

AREA	COMMANDS
System	/status /engine status /inspect context
Grounding	/web /web plan /web status /web auto on off /web watchlist
Memory	/memory facts /memory domains /memory timeline [domain] /memory audit
Private memory	/memory sensitive on off retention N /memory event forget purge
Response	/brief /normal /deep /create /response mode auto
Reports	/report now /report status /report sensitive on off
Workspace	/workspace /files /task status /approvals /approve /cancel
Sessions	/new /sessions /session /export /clean

First acceptance prompts

```
17:17 on the clock. Explain its numerology and show the exact digit sum.  
  
/web plan What does 17:17 mean spiritually?  
/web plan How do current Textual workers behave?  
  
/create Write a two-minute Project Intermix video script and preserve Cognition v1.4.0 and 8000 exactly.  
  
/workspace Create cognition_probe.py that prints COGNITION_OK, run it, verify the output, and complete the task.  
  
/report now
```

Expected signals

- 17:17, digit sum 16, digital root 7; no invented repeated digits.
- Spiritual interpretation stays local; current Textual behavior searches.
- Creation receives the 4K/4K split; workspace requires real verification.
- Report files appear under Workspace Lens and omit private text by default.

11 / INSTALLATION AND RECOVERY

A staged upgrade with a full rollback point

The installer changes code and adds SQLite objects only after the test and capability gates pass.

STAGE	CHECK OR ACTION
1	Run all isolated Cognition, Continuity, memory, resident, TUI, and web tests
2	Verify SQLite FTS5, Textual widgets, LiteRT API, and at least 8 GiB free storage
3	Copy current engine and memory plus version/launcher to pre_cognition_install_
4	Install every Python module using per-file temporary replacement; compile installed engine
5	Initialize additive schema v2 and print preserved counts
6	Run non-mutating legacy migration report; do not launch Sovereign

Install from Android Internal storage

```
cognition_zip="$HOME/storage/shared/Project_Intermix_Cognition_v1.4.0.zip"
test -f "$cognition_zip"
installer="$HOME/intermix-cognition-installer-$(date +%Y%m%d_%H%M%S)"
mkdir -p "$installer"
python -m zipfile -e "$cognition_zip" "$installer"
cd "$installer/Project_Intermix_Cognition_v1.4.0"
test "$(cat VERSION)" = "1.4.0"
bash ./install.sh "$HOME/project-intermix"
```

Recovery discipline

- Close Sovereign before install or restore.
- Keep the printed rollback path until several days of normal use pass.
- Do not clear Termux app data, delete sovereign.db, or delete the model as a first-line fix.
- The release does not modify the Debian Kokoro installation or Resonance files.

12 / EXPERIMENTAL ROADMAP

Intermix Observatory - isolated AI-to-AI stress lab

The AI-only network idea is documented as a separate experiment, not activated by Cognition.

Likely reference

Moltbook describes itself as a social network for AI agents where humans can observe. A Moltbook-like connector or local Private Agent simulation could reveal long-horizon consistency, compaction quality, lore drift, and prompt-injection behavior.

RULE	OBSERVATORY DEFAULT
Isolation	Separate observatory.db, session identity, report directory, and quotas
Authority	No workspace, private memory, credentials, shell, voice, or core memory writes
Network	Local simulation first; live connector read-only; no autonomous posting
External text	Always untrusted; never interpreted as tool or policy instruction
Memory	Immutable turns plus hash-linked epoch summaries; preference/lore proposals require review
Runtime	One conversation, bounded turns/epochs/time/storage, hard kill switch
Evaluation	Numbers/names, contradiction, recall precision, repetition, injection, latency, RAM, storage

Staged progression

- Local multi-persona simulation with no network.
- Sanitized replay corpus and deterministic evaluation.
- Read-only public observation.
- User-reviewed reply drafts submitted manually outside Intermix.
- A bounded connector only after privacy, identity, shutdown, and injection tests pass.

Core boundary

Generated preferences and lore remain reviewable proposals with supporting turn IDs. They may enrich an experimental persona, but they cannot redefine the user's preferences, expand tool authority, or enter private personal memory automatically.

13 / FEEDBACK AND REFERENCES

What to measure before reporting upstream

The included Google feedback draft is intentionally local and must be reviewed and submitted manually.

Measured fields

- Cold and warm initialization, prefill, first-token, decoding, and total latency.
- Selected input/output tokens, available memory, peak pressure, unload behavior, and Signal 9 reproduction rate.
- Numeric corruption and repair rate; retrieval precision; irrelevant sensitive recall rate.
- Provider success, citation quality, stale-result rejection, and request count.
- Workspace repair, cancellation, restart continuity, and storage growth per 100 turns.

Primary references

- LiteRT-LM: <https://github.com/google-ai-edge/LiteRT-LM>
- Brave Search API: <https://api-dashboard.search.brave.com/documentation>
- Tavily Search: <https://docs.tavily.com/documentation/api-reference/endpoint/search>
- Exa Search: <https://exa.ai/docs/reference/search>
- SearXNG Search API: https://docs.searxng.org/dev/search_api.html
- GitHub REST API: <https://docs.github.com/en/rest>
- PyPI JSON API: <https://docs.pypi.org/api/json/>
- Crossref REST API: <https://api.crossref.org/>
- Google Custom Search overview: <https://developers.google.com/custom-search/v1/overview>
- Moltbook: <https://www.moltbook.com/>

Release evidence

Bundle result: the 44-test suite is clean; 42 tests pass in the authoring environment and two Textual interaction checks are intentionally deferred to the Pixel installer where Textual 8.2.8 is verified. The complete device sequence is in DEVICE_TEST.md.

End of handoff